ANSWER KEYS - PROFESSOR COPY

Week 2 Lecture Quizzes and Final

Freshman Seminar — Week 2 (pass mark 6/8)

1. B - If everything after it still makes sense, it's a detail. If the lecture falls apart without it, it's load-bearing — a big idea.

2. C - The 78% is a supporting number and the bus is a supporting picture. The idea is what they support: most of the air is nitrogen and the body ignores it.

3. A - Professors state the idea, explain it, and state it again — so headings, section edges, and repeats are where to look. Tables and dates are usually details.

4. D - About 60% versus about 40%. Pulling an idea out of your own head (retrieval practice) builds the path; rereading only puts the words in front of your eyes.

5. A - His notebook page 'The Fetus in the Womb' came five hundred years before ultrasound. He drew it, asked a question about it, and wrote the idea in one line.

6. B - Keep the numbers a doctor would need at the bedside. Blood's pH window is one; the year and the brewery are details you can look up.

7. D - Write first, check second — that's retrieval practice. Next week the boxes only mark the spot; by Week 5 they're gone.

8. D - Forget '70–85%' and the lesson still makes sense — it's a supporting number. Forget any of the other three and something important stops making sense.

Chemistry 101 — Week 2 (pass mark 6/8)

1. B - An acid drops off loose hydrogens (H) like a bus dropping passengers. Those loose hydrogens are what your tongue tastes as sour — and what warps proteins if there are too many in the blood.

2. D - 7 is neutral, like pure water. Below 7 is acid; above 7 is base. Blood sits just above neutral, at 7.35–7.45.

3. C - Three steps down, and each step multiplies by ten: $10 \times 10 \times 10 = 1{,}000$. That's why a ruler from 0 to 14 can cover everything from bleach to stomach acid.

4. C - 7.35 to 7.45 — a window one-tenth of a step wide. Proteins are shaped to work right there, and they warp if the blood drifts.

5. B - CO_2 + water = carbonic acid (H_2CO_3), the same acid that makes soda water sharp. Every cell makes CO_2, so the lungs must keep blowing the acid away.

6. A - Low pH plus high CO_2 means carbonic acid is piling up because breathing isn't blowing it off. That's a breathing problem, so the fix is breathing help: more CPAP, ventilator breaths, or caffeine.

7. B - Bicarbonate is the blood's sponge — a base that catches loose hydrogens. Baking soda is sodium bicarbonate, which is why it fizzes against vinegar.

8. A - Sørensen worked at the Carlsberg brewery's laboratory in Copenhagen. Brewers needed an exact number for acid, so he invented one — and hospitals have used it ever since.

Biology 101 — Week 2 (pass mark 6/8)

1. A - Upstairs atria receive; downstairs ventricles pump. Two of each, because the heart is really two pumps side by side.

2. C - Right heart -> lungs -> left heart -> body -> right heart. The right side runs the lung loop.

3. D - 140 × 60 = 8,400 an hour; 8,400 × 24 ~ 200,000 a day. Twice what an adult heart does.

4. A - Red cells are bags of hemoglobin — the no-nucleus cells from Week 1. Plasma carries sugar, salt, and bicarbonate; white cells defend; platelets plug leaks.

5. B - About 80 mL for every kilogram. That's why every blood test is counted: 2 mL from an 80 mL baby is 1 part in 40.

6. D - Before birth the lungs aren't breathing, so the ductus lets blood detour around them. At birth it should squeeze shut within a few days.

7. A - Blood pumped toward the body sneaks back through the open pipe into the lungs. Breathing gets harder and the gut, kidneys, and brain get shortchanged.

8. C - He multiplied blood per beat by beats per hour and got far more than a body could make — so it had to circulate. Then he showed the valves with the finger trick from today's lab.

Physics 101 — Week 2 (pass mark 6/8)

1. D - The prism bent each color by a slightly different amount, fanning the beam into a rainbow. White light was never one thing — it was all the colors travelling together.

2. C - A thing's color is the light it refuses. The tomato soaks up blue, green, and the rest, and only the red comes back out.

3. A - Infrared has even longer waves than red. It's invisible, but you feel it as heat — and the pulse oximeter uses it as its second color.

4. A - Oxygen-loaded hemoglobin lets red through and absorbs infrared; empty hemoglobin does the opposite. The ratio of the two is the SpO_2 number.

5. D - Skin, bone, and vein blood absorb light too, but they don't pulse. Keeping just the beat-by-beat wiggle isolates the fresh arterial blood — that's the 'pulse' in pulse oximeter.

6. B - A yellow molecule absorbs blue. That energy twists bilirubin into a water-soluble shape that leaves in urine and bile — no liver enzyme needed.

7. C - A diaper-shaped patch of yellow was the clue. Light was clearing bilirubin through the skin — and Dr. Cremer built the first lamp from that observation, published in 1958.

8. C - The shields protect developing eyes from the brightness. (Ordinary NICU lighting was tested and does NOT cause ROP — oxygen does. The shields are about the bili lamp's intensity.)

Calculus 101 — Week 2 (pass mark 6/8)

1. C - Total = rate × time. 4 × 24 = 96 mL. 'Per hour' tells you to count hours.

2. B - A flat rate of 3 for 24 hours is a rectangle 3 tall and 24 wide — area 72, the total. Steepness was last week's question (how fast); area is this week's (how much).

3. D - Two rectangles: 2 × 6 = 12, plus 4 × 18 = 72. Add them: 84 mL.

4. B - Every 3 hours is 8 feeds a day (24 ÷ 3). 15 × 8 = 120 mL.

5. A - Divide the day's total by the weight: 150 ÷ 1 = 150 mL/kg/day. Around 150–160 is 'full feeds' for a growing preemie.

6. B - That's the slicing idea — Archimedes' polygons, Leibniz's stretched-S. Thin enough slices add up to exactly the area under the curve.

7. D - Wet weight minus dry weight, in grams, equals milliliters of pee — because a milliliter of water weighs one gram. Output is a total that piles up, just like intake.

8. C - Hexagon, then 12, 24, 48, 96 sides, inside and outside the circle — the slicing idea done with polygons. It got him pi to better than three decimal places.

Statistics 101 — Week 2 (pass mark 6/8)

1. A - Line them up first: 120, 130, 140, 150, 160. The middle one is 140. (The mean happens to be 140 too — that's not always so.)

2. B - Change the biggest baby from 1,600 g to 3,000 g and the mean jumps, but the fifth baby in line is still the fifth baby in line. The median only cares about order.

3. A - A percentile is a place in line out of 100. Tenth from the smallest end: about 10 in 100 are at or below her, about 90 are bigger.

4. B - Fiftieth in a line of a hundred is the middle — the median. Half the group is below, half above.

5. C - Position ÷ group size × 100: 7 ÷ 10 = 0.7, so the 70th percentile. Seven of ten babies are at or below her.

6. B - Below the 10th line for the week you were born is 'SGA' — a flag, not a diagnosis. The team looks for causes like a placenta that wasn't delivering well.

7. C - A percentile is a place in line, not a grade. One baby in twenty is at or below the 5th by definition. What doctors watch for is a change — crossing curves downward.

8. D - Tanis Fenton's 2013 chart pooled about four million births. The honest catch: they were babies who were BORN early — often because something was wrong — so a different chart built from healthy pregnancies sits a little higher.

Genetics 101 — Week 2 (pass mark 6/8)

1. B - Long enough for feeding to start and any pile-up to begin, but early enough to treat before harm. NICU babies get it repeated if TPN or transfusions could muddle the result.

2. B - Rich in blood, safely away from the heel bone. The poke is still real, so it's paired with sugar water, swaddling, or a parent's chest.

3. D - Gene -> enzyme -> job. In PKU the enzyme that breaks down phenylalanine doesn't work, so it piles up. The baby looks perfect at birth — which is why the screen exists.

4. B - A tiny worker with one skill, built from a gene's recipe. Thousands of kinds: some burn sugar, some digest dinner, one clears bilirubin.

5. C - Guthrie was a bacteria scientist. His strain feasted only when phenylalanine was high — so a ring of growth meant PKU, days after birth and before any harm.

6. A - Screen everyone when all four line up: the disease hides, waiting is a disaster, early treatment works, and the test is cheap and safe. For PKU, they do.

7. D - About 1 in 3,000 babies, versus 1 in 12,000 for PKU. A growth-and-brain hormone runs short; one small pill a day replaces it.

8. A - Tiny earphones and a sensor check hearing; a pulse ox on the right hand and a foot at 24 hours looks for the oxygen mismatch of a heart built with a missing pipe.

Psychology 101 — Week 2 (pass mark 6/8)

1. A - Bowlby argued it's a survival system as basic as hunger, not a side effect of being fed. Harlow's monkeys proved him right.

2. C - About 17 hours a day on the soft one — and they ran to it when frightened. Comfort beat food. (The experiments were also cruel, and wouldn't be allowed today.)

3. B - Upset when she leaves, comforted fast when she returns, then back to the toys: the rope is strong and the baby knows it. About six in ten babies.

4. B - Babies expect serve-and-return and work hard to get it back. When the mother's face returns, so does the baby — within seconds.

5. C - When she looked at the families, the difference wasn't in the babies; it was in how reliably someone came when the baby signaled.

6. C - From the 1940s to the 1970s, parents looked through glass a few times a week. Then Klaus and Kennell measured what early contact did, and the doors opened.

7. A - More than 3,000 babies in five countries. Not waiting until the baby was 'stable' saved lives. Bowlby's rope is a treatment with a death rate attached.

8. D - It's measurable — that's what makes it medicine. Which is why the reclining kangaroo chair is on the care plan and parents are on the team.

Sociology 101 — Week 2 (pass mark 6/8)

1. A - More than one in ten births. About 900,000 children a year die from it — the number-one cause of death in children under five.

2. D - A country with ten times the births will have ten times the deaths even if it's doing equally well. 'Per' means divide — the Calculus word — and dividing puts everyone on the same line.

3. C - Two regions, most of the babies — and most of the deaths, because basic care is hardest to reach there.

4. C - Not exotic things — the ordinary NICU Tutor things. The same baby, born in two places, gets two different lives. That's why kangaroo care and plastic wraps save so many.

5. B - About 10.9 per 1,000 versus about 4.5. Black babies are also born preterm far more often — about 15 in 100 versus 9 in 100.

6. A - More than two million American women of childbearing age live in one. Regionalization only works if the Level III unit is reachable.

7. B - Circumstances, not biology. 'Your ZIP code predicts your health better than your genetic code' — and change the circumstances, and the numbers move.

8. D - The Aspiranto Health Home: beds upstairs, pay what you can. Then Yale, public health, and reports the city couldn't ignore. Counting who isn't getting care is where change starts.

Week 2 Final Exam (pass mark 17/24)

1. D - Use the 'forget it' test: forget the walnut, the king, or the count of cord vessels and the lecture still stands. Forget the two-pump idea and none of the rest makes sense.

2. C - Chemistry: 7.35–7.45. Physics: oxygen-loaded hemoglobin is bright red and lets red light through. Biology: about 80 mL per kilogram — which is why every test's drops are counted.

3. B - Low pH plus high CO_2 points at the lungs. Low pH with NORMAL CO_2 would point at the cells. High pH with low CO_2 would mean over-breathing.

4. D - Three steps on Sørensen's ruler, ten times each: 10 × 10 × 10 = 1,000.

5. A - Right heart -> lungs -> left heart -> body -> right heart. The left ventricle has the thickest wall because it pushes the farthest.

6. C - Patent ductus arteriosus: 'patent' just means open. Before birth it let blood skip the unused lungs; after birth it's supposed to squeeze shut within days.

7. A - Loaded hemoglobin lets red through and absorbs infrared; empty hemoglobin does the reverse. The ratio is SpO_2 — measured on the pulsing part of the signal, Aoyagi's trick.

8. B - A yellow molecule absorbs blue. That's why the lamp works right through the skin, and why more skin showing means a faster cure.

9. B - Total = rate × time: 2.5 × 8 = 20 mL. (2 × 8 = 16, plus half of 8 is 4.)

10. B - Height × width = rate × time = total. A staircase is several rectangles; a curve is many thin ones. Steepness was last week's question.

11. C - In order: 820, 880, 900, 950, 1,000, 1,300, 2,400. The fourth of seven is 950. The mean (about 1,179) got dragged up by the 2,400-gram baby; the median didn't.

12. A - A percentile is a place in line, not a grade. What matters is CHANGE: crossing curves downward means the growth line has gone flat. Read the line, not the dot.

13. A - Gene -> enzyme -> job. A broken worker means a pile-up. Caught by the heel prick at 1–2 days and prevented with a special diet.

14. A - Four things line up, and screening makes sense. A positive screen is a wide net — 'test again, properly, now' — not a diagnosis.

15. B - About 17 hours a day on the cloth mother, and straight to it when scared. Attachment isn't a side effect of being fed.

16. D - Smiles, sounds, pointing, then fussing and looking away — and back to normal in seconds when the game resumes. Every return is a brain connection.

17. C - Counts and rates answer different questions. Divide by the group: A is 200 ÷ 2,000 = 10 per 100; B is 300 ÷ 6,000 = 5 per 100.

18. D - The same baby, born in two places, gets two different lives — and change the conditions (kangaroo care, wraps, a clinic) and the numbers move. It isn't the babies.

19. C - Torricelli's tube, 1643: the air holds up 760 mm of mercury. A blood pressure of 60 means the heart pushes hard enough to lift a 60 mm column.

20. A - 7–10 doing well; 4–6 may need help; 0–3 needs help now. And Apgar herself warned: it predicts the next few minutes, not a life.

21. B - About 20,000 recipes. This week you added one link: many of those proteins are enzymes, and a typo in an enzyme's recipe means a pile-up.

22. C - Servo control: the machine chases a target hundreds of times a day so the baby doesn't have to. NICU Tutor Lesson 1.

23. D - The espresso molecule, in a measured daily dose. Steady breathing also keeps CO_2 — and carbonic acid — from piling up, as this week's Chemistry showed.

24. D - The throat is a railroad junction where food and air cross. Until the rhythm clicks in, milk goes in by tube — and the pump's mL per hour was this week's Calculus.